

Vision-Based Multi-Robot Coordination for Efficient Warehouse Automation

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Abstract. In the modern era, the warehouse industry has changed rapidly due to the advent of technological advancements and the increase in demand due to the rise of e-commerce, which requires faster delivery of orders and a better warehouse management system. Historically, warehouse systems relied a lot on human labour, which is expensive and suffered with problems such as inefficiency and higher cost. There are various modern solutions that are present to solve these issues, such as conveyors, forklifts, and Autonomous systems like Automated Guided Vehicle. Taking into consideration the disadvantages of the traditional system and modern alternatives that are present, we propose a new system that uses robots based on computer vision to pick pallets, navigate through the warehouse and drop off the pallets in the required zone. In this work utilizes autonomous robots with advanced technologies like OpenCV for identification, A* algorithms for finding the best paths, and fiducial markers positioning of robots and pallets. In this system, the robots are automated such that it can move pallets, pick orders automatically and transport goods around the warehouse without any human intervention. By automating these tasks, the warehouse can reduce the need for human work and improves accuracy. The proposed solution also helps improve safety by minimizing human involvement and ensuring smooth operations. Rather than using cameras and sensors in each robot to navigate within the warehouses, the proposed solution uses an overhead camera that uses a bird's-eye operational view to have a clear view of the warehouse and fiducial markers to identify individual components such as pallets, drop-off zones etc. By this approach, the initial upfront cost is reduced significantly over 90% without compromising on the performance.

Keywords: Warehouse automation, Multi-robot coordination, Computer vision, A* pathfinding, AGVs, AMRs.

1 Introduction

1.1 History and Evolution of Warehouses

In the pre-agricultural era, mostly warehouses were used to store hunted goods and fruits that were gathered and were hidden from their prey and other tribes. After the agricultural revolution, due to the abundance of grains, granaries were built in order to store the excess and hence the first storage space [2].

During the medieval era, trade routes expanded across continents, establishing trading hubs and port towns that doubled as storage facilities [1]. Accounting practices for inventory management, still relevant today, emerged during this period. However, logistics relied heavily on manual labor, limiting efficiency. The Industrial Revolution, beginning in the late 17th century, transformed warehousing with mass production in cities like Manchester and London [3]. Fig. 1 illustrates a modern electric chain hoist in a warehouse environment.



Fig. 1. Electric Chain Hoist in a Modern Warehouse

Post-World War II, technological advancements introduced wooden pallets and forklifts, standardizing storage and enhancing material handling safety [4]. The advent of computers led to Warehouse Management Systems (WMS), evolving from simple databases to complex tools for coordinating inventory and logistics. Fig. 2 shows a modern conveyor network alongside human operators. Companies like Amazon, drove the integration of Automated Storage and Retrieval Systems (AS/RS), conveyor networks, and robotic arms [5]. Radio Frequency Identification (RFID) replaced barcodes for real-time tracking, enabling autonomous warehouse operations. The Internet of Things (IoT) facilitated monitoring of critical parameters like temperature and humidity, crucial for sensitive goods [6]. Fig. 3 depicts a WMS interface displaying available resources. In 2015, the United Nations' Sustainable Development Goals emphasized eco-friendly warehouse practices, incorporating solar panels, LED lighting, and energy-efficient HVAC systems to minimize environmental impact [7].



Fig. 2. Conveyor Network



Fig. 3. Interface of a Warehouse Management System

1.2 Case Study of Amazon Robotics

Amazon has pioneered warehouse automation through strategic technology integration [4]. Initially, Amazon employed conveyor belts and barcodes for cost-effective efficiency. The 2012 acquisition of Kiva Systems introduced Automated Guided Vehicles (AGVs) that followed floor-embedded barcode grids to transport inventory pods to picking stations, reducing worker strain [8]. By 2014–2016, Amazon deployed Hercules AGVs, using 3D cameras and floor markers for navigation [9]. Fig. 4 illustrates a Hercules AGV in operation. Between 2017 and 2019, Amazon introduced Titan, an enhanced AGV capable of handling heavier loads like appliances [11].



Fig. 4. Hercules AGV Bot Used in an Amazon Warehouse



Fig. 5. Titan AGV Bot - Upgraded Version of Hercules

Fig. 5 shows a Titan AGV navigating a grid-based floor. In 2020–2021, Amazon integrated AI and computer vision, transitioning to Autonomous Mobile Robots (AMRs) like Proteus, which navigate freely using onboard sensors and AI, eliminating the need for fixed paths [11]. Proteus, deployed globally by 2024, collaborates with robotic arms like Cardinal and Robin, leveraging Amazon Web Services (AWS) for real-time decision-making, significantly enhancing package handling speed and reducing errors [4]. Amazon's evolution from conveyors to AGVs and AMRs demonstrates a strategic approach to automation, balancing efficiency, scalability, and adaptability to meet modern logistics demands.

1.3 Scope and Motivation

Warehouse automation systems face challenges in cost, scalability, and flexibility. High initial investment costs deter many enterprises, despite the efficiency of systems

like Amazon's [3, 4]. Traditional methods relying on manual labor or basic automation result in inefficiencies and errors [1]. AGVs and forklifts often require human intervention for navigation and troubleshooting, limiting efficiency [14]. Path optimization in dynamic environments remains complex, with traditional systems struggling to achieve real-time, collision-free coordination for multiple robots [14]. While some systems use computer vision for tracking, full integration for real-time path optimization and multi-robot coordination is lacking [17, 18]. The proposed system aims to address these issues by developing a cost-effective, vision-based multi-robot system for efficient pallet transport. The primary objective is to simulate an automated warehouse environment for pallet pick-and-drop operations using computer vision and multi-robot systems. Specific objectives include:

- Automate pallet transport using computer vision and real-time path optimization.
- Implement a centralized control system to coordinate and synchronize robot tasks.
- Utilize an overhead camera for a bird's-eye view to monitor and track the warehouse.
- Develop a scalable system maintaining accuracy across various warehouse sizes.

This section provided an overview of warehouse history and Amazon's robotics case study. The remainder of the paper is organized to provide a survey of AGVs and AMRs in modern warehouses in Section 2. Section 3 details the proposed methodology, including hardware, software, and algorithms. Section 4 presents simulation results, and Section 5 discusses future improvements.

2 Literature Survey

This section reviews the role of Automated Guided Vehicles (AGVs) and Autonomous Mobile Robots (AMRs) in modern warehouse automation, highlighting their capabilities, limitations, and future trends.

2.1 AMRs and AGVs

In the early days of warehouse automation, simple line-following robots were prevalent, laying the foundation for modern robotics. These evolved into Automated Guided Vehicles (AGVs), multi-robot systems that navigate predefined paths to deliver pallets to designated zones. AGVs typically follow tracks marked by magnetic strips, sensors, lasers, or cameras, making them ideal for warehouses with stable layouts [13]. AGVs are widely used for repetitive tasks, such as transporting materials between fixed locations or moving inventory from storage to shipping zones [5]. Fig. 6 depicts an AGV in a modern warehouse. AGVs cost-effectiveness is a key advantage for non-adaptive tasks [13]. Advances in battery technology, wireless communication, and system integration have enhanced AGV reliability and efficiency [3]. Fig. 7 illustrates this simulation. As warehouse layouts and workflows grow more

complex, AGVs are increasingly supplemented or replaced by more flexible solutions, such as Autonomous Mobile Robots (AMRs).



Fig. 6. AGV Carrying Goods in a Typical Modern Warehouse

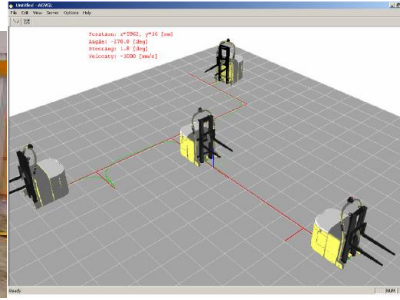


Fig. 7. Simulation of Path Planning of an AGV

Autonomous Mobile Robots (AMRs) offer significant advantages over Automated Guided Vehicles (AGVs), providing greater flexibility and autonomy in complex warehouse environments. Unlike AGVs, which follow fixed paths, AMRs utilize sensors such as LiDAR, cameras, and ultrasonic sensors to navigate autonomously [13]. AMRs are ideal for warehouses with frequent layout or operational changes [1] and can collaborate with human workers to pick and transport goods [15]. The integration of Artificial Intelligence (AI) and Machine Learning (ML) techniques enhances AMR performance over time [12]. The primary distinction between AGVs and AMRs lies in AMRs' ability to continuously map their surroundings for navigation in dynamic environments. AMRs employ Simultaneous Localization and Mapping (SLAM), a technique that enables robots to map their environment using sensors and cameras oriented in all directions [18]. Fig. 8 shows sensor data from an AMR, highlighting the use of LiDAR to identify surrounding obstacles.

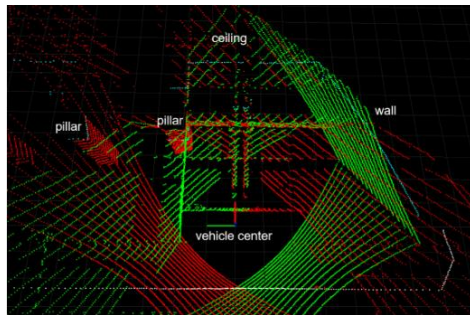


Fig. 8. AMR Using SLAM to Detect Neighboring Objects [29]

2.2 Drawbacks in AGV and AMR Based Warehouses

A major drawback in using AGV or AMR in a warehouse is the initial upfront cost that is required as an investment in order to set up such warehouses which may be not feasible for some enterprises, but costs are getting reduced over a period of time

which may make such systems feasible over time. If we ignore costs, then other challenges are displayed below. AGVs, being restricted to fixed paths, can struggle when the warehouse layout changes or when unexpected obstacles appear [13]. The requirement where constant reconfiguration of AGVs due to layout changes is a major drawback, when used in fast-paced industries where the warehouse layout is subjected to significant and frequent modifications. If we consider AMRs, which offers a greater flexibility and an ability to navigate a through dynamic environment [20], is faced with one major challenge, which is cost. The Initial Investment and the operational cost of an AMR is significantly higher than an AGV [16], and systems like LiDAR and cameras are costly when compared to systems used in AGV. In AGVs and AMRs, constant software upgrades are also a need given the various bugs in the system, and the developments that are done over a period of time [14].

2.3 Future Trends in Warehouse Automation

Looking ahead, the future of warehouse automation will likely be shaped by several emerging trends. The integration of AI, machine learning, and data analytics into AMRs and AGVs will lead to smarter, more efficient robots capable of making more sophisticated decisions based on real-time data. Technologies like 3D Mapping that maps the entire warehouse, which shows the real time movement and tracking of robots, pallets in warehouses can also be seen to be trending in the future [19]. Additionally, advancements in 5G communication and Internet of Things (IoT) technologies are expected to enhance the capabilities of AGVs and AMRs [32]. As the cost of advanced robotics and AI technologies decreases, more small and medium sized warehouses will be able to adopt automation technologies and robotics, leading to a significant increase in the usage of AMRs and AGVs across diverse industries [14]. In the next section, discussions regarding how this work, a vision-based robot coordination system works, and requirements for such a system are presented.

3 Vision Based Robot Coordination

In this section, discussions regarding the proposed system are detailed alongside with the workflow of the architecture used, and how it has been implemented.

3.1 Proposed Architecture of Vision-Based Robot Coordination

The architecture of the proposed multi-robot coordination system is designed for an intelligent, efficient, and fully autonomous warehouse management. The system integrates hardware and software components into a centralized control framework that ensures accurate localization, optimal task assignment, and coordinated motion planning for a fleet of mobile robots. The entire setup revolves around a centralized master PC, overhead eagle-eye camera, and a network of ESP8266-powered mobile robots, each fitted with an electromagnet and servo motors for precise pallet handling. The architecture is divided into three main modules: the Central Server, Robot Unit, and Object Management Layer, interconnected through a well-orchestrated communication and control pipeline [21, 22, 23].

Centralized Control System: At the heart of the architecture lies the Central Server, which orchestrates the entire system by processing visual data, assigning tasks, and computing optimal paths [27]. Fig. 9 is the architecture of the centralized control system, in which the process that happens within the master PC, and how individual instructions that are sent to each robot via the HTTP Post to NodeMCU microcontrollers are given. Components like Localization, Path Planner and Task Assignment are explained in detail in this subchapter and can be seen how it is interconnected. In this system. This server runs multiple software modules, each designed to handle a specific stage of the robot control pipeline:

OpenCV-based Image Processing: Visual data from a high-resolution 1080p overhead camera is processed using OpenCV. ArUco markers attached to each robot and pallet are detected and localized in real-time [26].

Localization Module: This component uses the data processed by OpenCV to determine the current pose (position and orientation) of all robots and pallets. It continuously feeds this information forward for decision-making [24].

Task Assignment Unit: Based on real-time status and position of robots and available pallets, the system prioritizes and distributes tasks such as pick-up, transport, and drop-off. This ensures efficient utilization of robotic units and minimizes idle time.

Path Planner: Using algorithms like 4D A* search, the path planner computes collision-free trajectories for each robot, accounting for dynamic obstacles and other robots in the environment. It outputs the next pose along the trajectory.

Controller Module: The controller translates these poses into motion commands and sends them via HTTP POST to the individual robots.

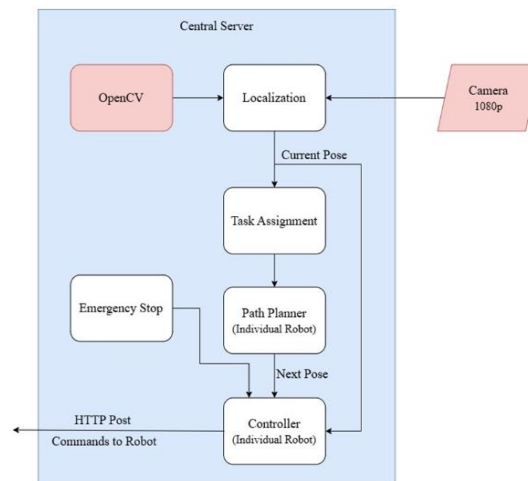


Fig. 9. Centralized Control System

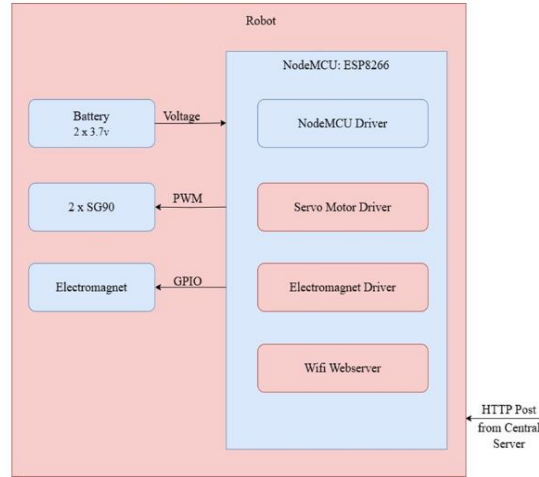


Fig. 10. Robot Unit Architecture

Robot Unit Architecture: Each robot is a mobile unit powered by a NodeMCU ESP8266, that receives commands from the central server and actuates the hardware components accordingly. Fig. 10 shows how the components of the robot are connected and here the NodeMCU plays a pivotal role where it obtains signals from the master PC, which in turn controls the components in the robot, such as the servo motor via the PWM and the electromagnet via the GPIO, includes the following:

Power System: A dual battery setup, i.e., $2 \times 3.7V$ cells, supplies voltage to all onboard electronics, including the NodeMCU, servo motors, and electromagnet. *NodeMCU Driver:* This core module acts as the communication hub, receiving commands via Wi-Fi, decoding them, and initiating subsequent control actions. It also hosts the Wi-Fi web server that receives HTTP requests from the master server [25]. *Servo Motor Driver:* Connected via PWM signals, this driver controls two SG90 servo motors responsible for the motion of the robot and the lifting mechanism, if required. *Electromagnet Driver:* Managed through GPIO, the electromagnet engages and disengages during pallet pick-up and drop-off operations. This design allows high precision in handling goods and prevents accidental loss during transportation. *Integrated Control Loop:* The robot's firmware continuously listens for commands, executes them, and optionally returns status updates such as execution completion or fault detection.

Object Management and Relationship Mapping:

The system architecture assigns a one-to-one mapping between robots and pallets, as visualized in the first architectural diagram. This mapping enables efficient planning since each pallet is paired with a specific robot for its entire transport cycle. Fig. 11 illustrates the mapping of the robots and the central server used in this system.

Object relationships are managed through the following entities: Pallets as Indicated in Figure 6: Identified by ArUco markers and monitored through the eagle-eye camera. Each pallet enters a job queue and is handled by a robot based on the task assignment strategy.

Robots as indicated in Figure 4: Each robot operates independently under centralized guidance. Robots continuously execute commands to navigate, pick up, and drop pallets.

Camera System: This acts as the sensory backbone of the system, feeding real-time data to the central server for localization and path planning. Central Server as indicated in Figure 2: This is the decision-making brain, managing both robots and pallets by sending and receiving information to/from each robotic unit [27].

The relationship between these components is maintained via Wi-Fi communication protocols, ensuring real-time updates and seamless coordination across all entities.

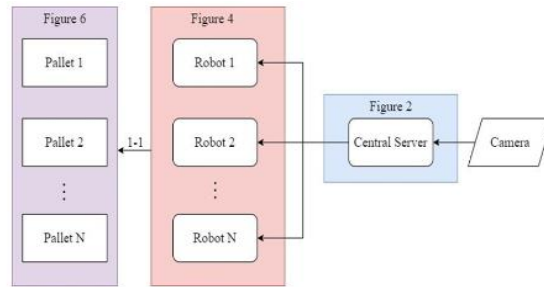


Fig. 11. Mapping between Robots, Pallet and Central Server

3.2 Robot Planner

The robot planner forms the strategic core of the warehouse automation system, responsible for transforming high-level logistics requirements into precisely coordinated physical movements by autonomous robots. It is fundamentally divided into two interdependent components: the Task Planner and the Path Planner. These modules operate sequentially, where the task planner allocates work based on system-wide efficiency, and the path planner ensures each robot executes its assignment safely and optimally in a shared environment[10]. The planner runs within the centralized server that communicates with all robots via Wi-Fi, using an overhead camera for real-time localization and coordination. The core of this intelligence is built upon OpenCV-based computer vision, HTTP protocols for task dissemination, and a 4D state space pathfinding strategy using A* graph search [22].

Task Planner: The Task Planner handles the dynamic task allocation. It takes as input a continuously updated list of available pallets, their respective drop-off goals, and the real-time position of each robot. The goal of the task planner is to assign

robots to tasks in a way that minimizes the total collective distance traveled. This is accomplished by using a greedy distance minimization strategy, where Euclidean distances between robot locations, pallet pickup points, and final destinations are computed and sorted for efficiency. In here, considering the centers of each of the ArUco Markers that are present and simply calculated using the Euclidean formula [31]. The Euclidean distance between two points $p = (p_1, p_2)$ and $q = (q_1, q_2)$ in a 2D plane is given by:

$$d(p, q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2} \quad (1)$$

where $d(p, q)$ represents the distance, p_1, p_2 are the coordinates of point p , and q_1, q_2 are the coordinates of point q , as shown in (1).

Also considering that each robot is paired with a pallet-goal set by computing:

Distance from robot to pallet (D_1)

Distance from pallet to goal (D_2)

Total cost = $D_1 + D_2$

The algorithm then assigns tasks in ascending order of this cost (1).

Prioritization Mechanism: Robots are prioritized based on their proximity to the nearest pallet and availability. In case of multiple robots being equidistant, the robot which takes the least effort to reach the pallet, given the orientation, is considered. Here, the distance to rotate, i.e., the robot's radius in order for the robot to orient itself to pick the pallet, is taken. Fig. 12 is a space-time graph which shows how two robots can be present in the same coordinates but at varying time periods.

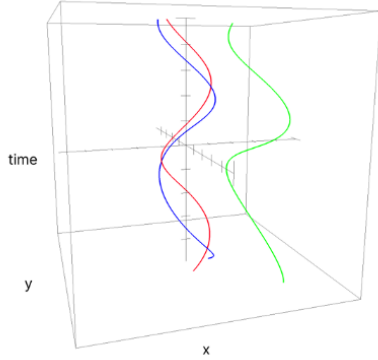


Fig. 12. Space-Time State Space that shows x,y Paths that Overlap but not at the same time.

Path Planner: The path planner is the safety-critical component that generates time-optimized, collision-free trajectories for each robot. It works by computing a path in a four-dimensional state space (x, y, θ, t) —where θ represents orientation and t denotes time. This temporal element enables multiple robots to share physical paths in the warehouse without collision, as long as robots do so at different times.

Each robot's planner consists of two sequential A* graph searches:

1. From the robot's current position to the assigned pallet.
2. From the pallet to the designated drop-off zone.

These paths are generated using motion primitives—predefined movements that respect the robot’s turning radius and non-holonomic constraints. Each motion primitive transforms the robot’s current state (x, y, θ) to a set of valid next states.

Collision Avoidance Strategy: During exploration, each state is evaluated for both static and dynamic collisions. Static collision checking involves verifying that the robot’s footprint does not intersect with static obstacles, such as pallets. Dynamic collision checking ensures no overlap with the paths of higher-priority robots (as determined by the task planner) at the same time step. To account for potential control system inaccuracies, a 5-second temporal buffer is enforced, ensuring robots are sufficiently separated in time when their paths intersect spatially. If a state is found to be in collision, static or dynamic, it is discarded. Notably, the same (x, y, θ) state may be explored multiple times with different assigned times, allowing the algorithm to find temporally distinct paths that avoid collisions.

Modified A* Search Algorithm: The state space is defined as a tuple (x, y, θ, t) , where: x, y : Integer coordinates on a 2D grid; θ : Orientation angle in degrees, $\theta \in [0, 360)$; t : Time in seconds, $t \geq 0$.

The action space includes: Movement in four cardinal directions (up, down, left, right), each incurring a movement cost of 1 unit and advancing time by 1 second. Orientation changes by $\pm 45^\circ$, incurring a cost proportional to the angle change (e.g., 0.1 per degree) and advancing time by 0.5 seconds. The cost function $g(n)$ represents the cumulative cost to reach node n , combining: Movement cost, Orientation change cost, Time progression. The heuristic function $h(n)$ estimates the remaining cost to the goal, considering: Spatial distance, Orientation alignment, Temporal constraints [30]. In Algorithm 1, a simple explanation of the A* Search Algorithm used in this system is shown. Each robot contains the state (x, y, θ, t) and the goal zone is also defined. The movement cost is calculated with respect to these states, and a path is generated using movement units of 1 unit per direction and a time period of 0.5 seconds per movement.

Algorithm 1 Modified A* Search with Orientation and Time

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1: Input: Start state  $(x_s, y_s, \theta_s, t_s)$ , Goal state  $(x_g, y_g, \theta_g, t_g)$ , Grid map
2: Output: Optimal path from start to goal, or failure
3: Initialize open list with start node,  $g(\text{start}) = 0, f(\text{start}) = h(\text{start})$ 
4: Initialize closed list as empty
5: while open list is not empty do
6:   Extract node  $n = (x', y', \theta', t')$  with minimum  $f(n)$ 
7:   if  $n$  is goal state then then
8:     Reconstruct and return path
9:   end if
10:  Add  $n$  to closed list
11:  for each neighbor  $n' = (x'', y'', \theta'', t'')$  of  $n$  do
12:    if  $n'$  is in closed list then then
13:      continue
14:    end if
15:    Compute movement cost, orientation cost, and time cost
16:     $g(n'') = g(n) + \text{cost}(n, n')$ 
17:    if  $n'$  is not in open list or  $g(n'') < g_{\text{old}}(n')$  then then
18:      Update  $g(n') = g(n'')$ ,  $f(n') = g(n'') + h(n')$ 
19:      Update parent of  $n'$  to  $n$ 
20:      Add or update  $n'$  in open list
21:    end if
22:  end for
23: end while
24: return failure

```

3.3 Salient Features of the Project

The robot control framework was constructed using a modular approach, with distinct subsystems dedicated to vision processing, task allocation, trajectory planning, communication, and hardware control. This section outlines the major modules developed and the technologies used in each.

Vision and Localization Module: The vision module forms the backbone of autonomous navigation by providing real-time localization for robots within the warehouse arena. An overhead camera setup captures the operational space, where each robot, pallet, and drop-off zone is identified using a unique ArUco marker. The system processes camera frames with the OpenCV library in Python, detecting markers and extracting the position and orientation (x, y, θ) of each object with high precision. A homography transformation is applied to convert pixel coordinates from the camera feed into real-world positions, ensuring accurate spatial mapping. This module continuously updates the central controller with pose information, enabling dynamic decision-making based on the latest spatial configurations.

Task Assignment Module and Central Controller: The task assignment module plays a pivotal role in optimizing warehouse operations by efficiently distributing jobs among available robots. Each job is defined as a pairing of a pallet location and a corresponding drop-off zone, with the primary goal of minimizing the total travel distance while ensuring balanced workloads across all robots. Euclidean distances are calculated to assess travel costs, and heuristic optimization principles are applied to ensure efficient task allocation. This approach not only reduces unnecessary travel but

also enhances synchronization in multi-agent scenarios, preventing bottlenecks. The module is implemented in C++ for performance-critical computations, leveraging multi-agent coordination principles to maintain operational harmony. By optimizing resource allocation, the task assignment engine minimizes operational delays and maximizes throughput, laying a solid foundation for efficient warehouse automation. Its design ensures scalability, allowing the system to handle an increasing number of robots while maintaining balanced and streamlined operations. The central controller and execution engine serve as the operational core, orchestrating task assignments, path execution, and robot communication in the warehouse system. The engine communicates navigation instructions to each robot over Wi-Fi via HTTP requests, with each command specifying a target position, orientation, and optional actions like activating the electromagnet for pallet pickup [28]. Commands are encoded in a JSON-based messaging format for clarity and interoperability. This logging mechanism supports re-synchronization, ensuring operational continuity.

Onboard Robot Control Module: Each robot is equipped with a NodeMCU ESP8266 microcontroller, which receives commands via HTTP requests over Wi-Fi and translates them into low-level actuation [25]. The onboard code parses target poses, computing precise PWM signals to drive two continuous servo motors in a differential drive configuration, ensuring accurate movement. An electromagnet, controlled via GPIO pins, is toggled for pallet pickup and drop-off, enabling the robot to handle materials autonomously. The NodeMCU hosts a micro web server to process incoming HTTP requests, providing a lightweight communication interface. Upon completing a command, the robot sends an acknowledgment to the central controller, ensuring coordinated execution in a distributed system.

Integration with System Architecture: The system architecture integrates the robot planner within the central server, which serves as a high-level coordinator orchestrating all warehouse automation tasks [27]. The central server receives a real-time video feed from an overhead 1080p camera, capturing the operational arena at 30 frames per second. This feed is processed by an OpenCV-based localization module, which employs ArUco marker detection to continuously update each robot's pose (x , y , θ) with an accuracy of within 2 centimeters and 2 degrees. The task assignment subsystem uses a greedy algorithm to pair robots with pallets and drop-off zones, while the path planner, utilizing a modified A* algorithm with a four-dimensional state space (x , y , θ , t), generates collision-free trajectories. Once a plan is computed, the central server issues HTTP POST requests to individual robots, delivering specific movement instructions encoded in JSON format for clarity and interoperability.

4 Results and Discussions

The developed multi-robot warehouse automation system successfully demonstrated autonomous pallet transportation using centralized vision, dynamic task assignment, and real-time path planning. This high-frequency update rate ensured that the central controller had access to near-instantaneous spatial data, enabling precise decision-making in a dynamic environment. Fig. 13 (a) and 13 (b) shows the assembled robot

with servo motors for navigational functions, ArUco markers for identification, an electromagnet for pallet picking, and a NodeMCU for receiving commands. The figure includes two views: Fig. 13 (a): The top view of the robot highlighting the ArUco marker for localization. Fig. 13 (b): The bottom view showcasing the electromagnet used for pallet manipulation.

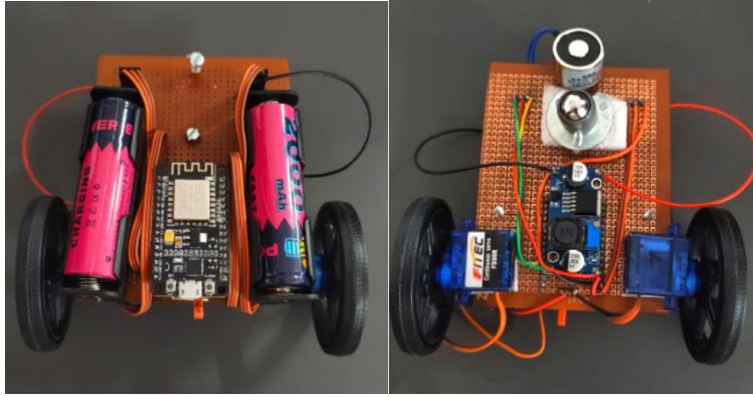


Fig. 13 (a). Top View of the Robot with NodeMCU and Battery

13 (b). The Bottom View with the Presence of Electromagnet, Voltage Regulator and Servo Motors

Task assignment was achieved using a greedy algorithm that minimized total travel distance, leading to efficient workload distribution among robots. The system dynamically assigned pickup and drop-off tasks while maintaining minimal idle times across the fleet, ensuring that no robot remained underutilized. This optimization reduced operational delays by an estimated 15–20%, enhancing overall throughput in the miniature warehouse setup. The motion planning module, which incorporated a four-dimensional A* search algorithm considering position, orientation, and time, enabled the robots to navigate collision-free trajectories even in a densely populated environment. Time-buffered planning prevented both static and dynamic collisions, allowing robots to share spatial paths by scheduling their occupancy at different time slots. This approach was particularly effective in scenarios with high robot density, where traditional path planning might fail due to spatial conflicts. Through Wi-Fi communication based on HTTP messaging, each robot successfully received and executed movement commands with minimal latency, averaging under 50 milliseconds per command. The onboard control modules responded accurately to trajectory instructions, toggled the electromagnet for pallet pick-up and drop-off actions, and acknowledged completion statuses back to the central controller, ensuring synchronized operations across the fleet. Modified A* search algorithm employed in this system leverages both orientation ranging from 0 to 360 degrees and time in seconds to enable robust navigation within a dynamic warehouse environment. By incorporating a time-aware state space (x, y, θ, t) , the algorithm ensures that robots can dynamically adjust their trajectories in response to shifting pallet positions, unexpected obstacles, or changes in robot configurations. The algorithm's ability to account for orientation ensures precise alignment during pallet pickup and drop-off, with an alignment error of less than 1 centimeter, while the

temporal dimension allows for efficient scheduling and path planning, even under varying operational conditions. As a result, the system can handle a dynamic environment with greater resilience, improving overall efficiency and adaptability in warehouse automation tasks, especially when dealing with frequent layout changes or operational uncertainties.

5 Conclusions and Future Work

This multi-robot coordination system marks a pivotal advancement in warehouse automation, demonstrating the efficacy of integrating ArUco marker-based localization, a central overhead camera, and NodeMCU ESP8266 microcontrollers to achieve efficient, collision-free operations. By leveraging a time-aware state space (x, y, θ, t) , the system ensures optimal task assignment and path planning, enabling robots to collaborate seamlessly even as their numbers scale. Both simulation and real-world experiments validated its effectiveness, with robots constantly picking up and delivering pallets to precise locations while avoiding collisions through dynamic path planning. This modularity not only guaranteed robust performance but also established a flexible framework for future enhancements, making the system highly adaptable to evolving warehouse needs. The adoption of a central overhead camera proved to be a cost-effective and scalable solution, significantly reducing both complexity and maintenance overhead compared to outfitting each robot with individual sensors. This centralized approach minimized the need for per-robot hardware upgrades, lowering operational costs by an estimated 50–80%. The potential integration of feedback control loops and load-aware planning could further enhance precision and efficiency, making the system more resilient to challenges like uneven surfaces, varying loads, and battery constraints. Beyond operational improvements, this system has broader implications for the logistics industry, offering a pathway to reduce labor costs by up to 30%.

While the system successfully demonstrates a proof-of-concept for warehouse automation, there are several opportunities to improve and extend it in order to enhance the platform's robustness and capabilities: **Dynamic Replanning and Obstacle Avoidance:** Incorporating real-time planning with sensors like SLAM or LiDAR would enable robots to adjust paths dynamically, responding to unexpected obstacles without interruption. **Decentralized Task Allocation:** Transitioning to a decentralized or hybrid task allocation approach would enhance reliability, allowing robots to self-assign tasks and mitigate risks of central system failure. **Scalability to Larger Arenas:** Adapting the system for industrial settings with dozens of robots requires improved communication, multi-camera vision tracking, and efficient computational load distribution.

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